Appendix A

Derivation of the Governing Differential Equations

A1.0 Introduction

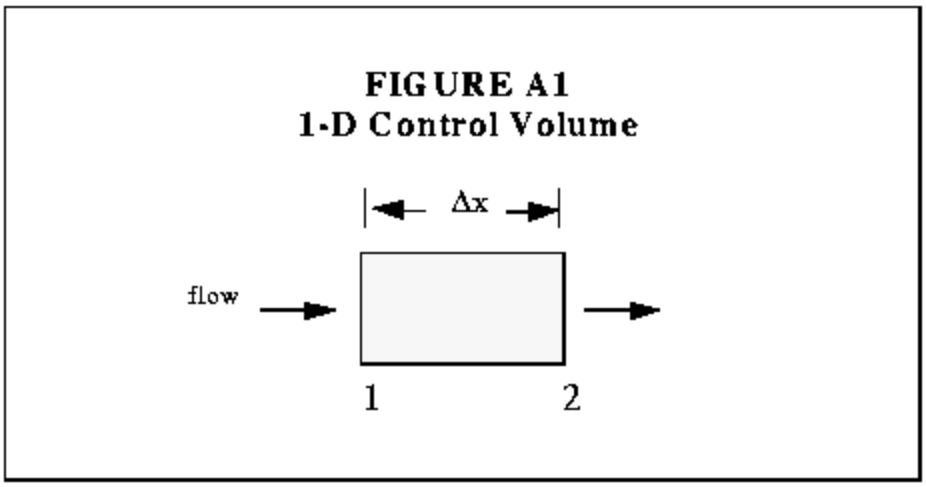
The primary purpose of this appendix is to derive the differential equations presented in Section 3.2. Implicit in this derivation are several underlying assumptions:

- The fate and transport of the modeled solute can be described by a one-dimensional system.
- Model parameters may be spatially and temporally variable. Spatial variation is assumed to occur in the longitudinal direction only.
- The mechanisms shown in Table 1 influence solute behavior in the stream channel and the storage zone, as indicated.
- The volume of the storage zones does not change in time.

Mechanistic models such as the one presented in this report are based on a fundamental engineering principle, *conservation of mass*. This principle states that for a finite volume of water, mass is neither created or destroyed (see Chapra and Reckhow, 1983). In short, mass conservation implies that:

$$Accumulation = Mass\ In - Mass\ Out \quad \text{(A1)}$$

Equation (A1) is often called a mass balance equation. This mass balance must be applied to a finite volume of water, known as a control volume. A control volume for our one-dimensional system is shown below. Note that the terms 'segment' (introduced in Section 3.2) and 'control volume' can be used interchangeably. In figure A1, Δx represents the segment length, while `1' and `2' are used to demarcate the upstream and downstream boundaries of the control-volume.



The remainder of this appendix is devoted to the formulation of differential equations based on Equation (A1). To accomplish this task, we first define `*Accumulation*' and then proceed to develop the `*in*' and `*out*' terms for each process given in Table 1. This procedure is carried out for both the stream channel and the storage zone.

A2.0 Mass Balance - Stream Channel

I. Accumulation

By definition, accumulation is the change in mass with respect to time. For the main stream channel, accumulation is given by:

$$Accumulation = \frac{\partial (VC)}{\partial t} \quad \text{(A2)}$$

where

- C in-stream solute concentration [M/L^3]
- V volume of the main channel segment [L^3]
- t time [T]

We can now expand Equation (A2) using the chain rule, yielding:

$$\frac{\partial (VC)}{\partial t} = V\frac{\partial C}{\partial t} + C\frac{\partial V}{\partial t} = \Delta x A\frac{\partial C}{\partial t} + \Delta x C\frac{\partial A}{\partial t} \quad \text{(A3)}$$

where

- Δx segment length [L]
- A stream cross-sectional area [L^2]

II. Advection

We now begin to develop the `*in*' and `*out*' terms for each process. The mass advected into the control volume, i.e. across the upstream boundary, is given by:

$$In = (QC)_1 \quad \text{(A4)}$$

where Q is the volumetric flowrate [L^3/T].

Next, the mass advected out of the control volume, i.e. across the downstream boundary, is given by:

$$Out = (QC)_2 = (QC)_1 + \frac{\partial}{\partial x}(QC)\Delta x \quad \text{(A5)}$$

Finally, we combine the `*In*' and `*Out*' terms, giving:

$$In - Out = -\frac{\partial}{\partial x}(QC)\Delta x = -\left(Q\frac{\partial C}{\partial x} + C\frac{\partial Q}{\partial x}\right)\Delta x \quad \text{(A6)}$$

III. Dispersion

In a similar fashion, we define the dispersive flux across the upstream and downstream boundaries:

$$In = -\left(AD\frac{\partial C}{\partial x}\right)_1 \quad \text{(A7)}$$

$$Out = -\left[\left(AD\frac{\partial C}{\partial x}\right)_1 + \frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right)\Delta x\right] \quad \text{(A8)}$$

$$In - Out = \frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right)\Delta x \quad \text{(A9)}$$

where D is the dispersion coefficient [L^2/T].

IV. Lateral Flows

Here we are interested in mass entering the system via lateral inflow, and mass leaving through lateral outflow. Note that unlike the previous two mechanisms, the concentrations associated with mass inflows and outflows differ (C_L vs. C):

$$In = C_L q_{LIN} \Delta x \quad \text{(A10)}$$

$$Out = C q_{LOUT} \Delta x \quad \text{(A11)}$$

$$In - Out = (C_L q_{LIN} - C q_{LOUT}) \Delta x \quad \text{(A12)}$$

where

C_L solute concentration of the lateral inflow [M/L^3]

q_{LIN} lateral inflow rate [L^3/T-L]

q_{LOUT} lateral outflow rate [L^3/T-L]

V. Transient Storage

The final physical mechanism we consider for the stream channel is transient storage. The inflow and outflow terms are given by:

$$In = \alpha V C_S \quad \text{(A13)}$$

$$Out = \alpha V C \quad \text{(A14)}$$

$$In - Out = \alpha V (C_S - C) = \alpha \Delta x A (C_S - C) \quad \text{(A15)}$$

where

α storage zone exchange coefficient [/T]

C_S storage zone solute concentration [M/L^3]

Note that the exchange coefficient, α, is defined as the fraction of the main channel volume that is exchanged with the storage zone, per unit time.

VI. Accumulation = In - Out

Now that the individual processes have been elucidated, the results from I-V may be used to develop the mass balance (Equation (A1)). Setting the accumulation term equal to the combined `*In - Out*' terms and dividing by Δx gives:

$$A\frac{\partial C}{\partial t} + C\frac{\partial A}{\partial t} = -\left(Q\frac{\partial C}{\partial x} + C\frac{\partial Q}{\partial x}\right) + \frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right) + (C_L q_{LIN} - C q_{LOUT}) + \alpha A(C_S - C) \quad \text{(A16)}$$

Although Equation (A16) represents a mass balance for a one-dimensional stream, it may be simplified using the water balance shown below.

VII. Water Balance

Like the solute mass balance presented above, a water mass balance is based on conservation of mass. Here again we develop Equation (A1) for a one-dimensional control volume or stream segment. Using the assumption of incompressible flow, accumulation is given by:

$$Accumulation = \frac{\partial}{\partial t}(V\rho_w) = \rho_w \Delta x \frac{\partial A}{\partial t} \quad \text{(A17)}$$

where ρ_w is the density of water [M/L^3].

The inflow term for the water balance is the sum of the water advected across the upstream interface and the water entering via lateral inflow:

$$In = Q_1 \rho_w + q_{LIN} \Delta x \rho_w \quad \text{(A18)}$$

Similarly, outflow is the sum of the water advected out and that leaving through lateral outflow:

$$Out = Q_2 \rho_w + q_{LOUT} \Delta x \rho_w = \left(Q_1 + \frac{\partial Q}{\partial x}\Delta x\right)\rho_w + q_{LOUT} \Delta x \rho_w \quad \text{(A19)}$$

Finally, the water balance is completed by combining Equation (A17) through (A19), and dividing by $\rho_w \Delta x$:

$$\frac{\partial A}{\partial t} = -\frac{\partial Q}{\partial x} + (q_{LIN} - q_{LOUT}) \quad \text{(A20)}$$

VIII. Governing Equation - Stream Channel

Substituting Equation (A20) into Equation (A16) and dividing by A yields the governing differential equation for the stream channel:

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right) + \frac{q_{LIN}}{A}(C_L - C) + \alpha(C_S - C) \quad \text{(A21)}$$

This equation is presented in Section 3.2.1 as Equation (1).

A3.0 MASS BALANCE - STORAGE ZONE

As in Section A2.0, a mass balance procedure is used to develop the governing equation. Here the approach is greatly simplified, as the processes of advection, dispersion and lateral inflow do not affect the storage zone.

I. Accumulation

Accumulation of solute mass in the storage zone is given by:

$$\frac{\partial}{\partial t}(V_S C_S) = V_S \frac{\partial C_S}{\partial t} + C_S \frac{\partial V_S}{\partial t} = \Delta x A_S \frac{\partial C_S}{\partial t} \quad \text{(A22)}$$

where

V_S volume of the storage zone segment [L^3]

A_S storage zone cross-sectional area [L^2]

Note that the storage zone area, A_S, is assumed to be constant in time.

II. Transient Storage

The change in storage zone solute mass is given by:

$$In = \alpha V C \quad \text{(A23)}$$

$$Out = \alpha V C_S \quad \text{(A24)}$$

$$In - Out = \alpha \Delta x A (C - C_S) \quad \text{(A25)}$$

where α is the fraction of the main channel volume that is exchanged with the storage zone, per unit time.

III. Governing Equation - Storage Zone

Combining Equations (A22) and (A25) we arrive at:

$$\frac{dC_S}{dt} = \alpha \frac{A}{A_S}(C - C_S) \quad \text{(A26)}$$

This governing equation is presented in Section 3.2.1 as Equation (2).

Appendix B

Finite Difference Approximations and Boundary Conditions

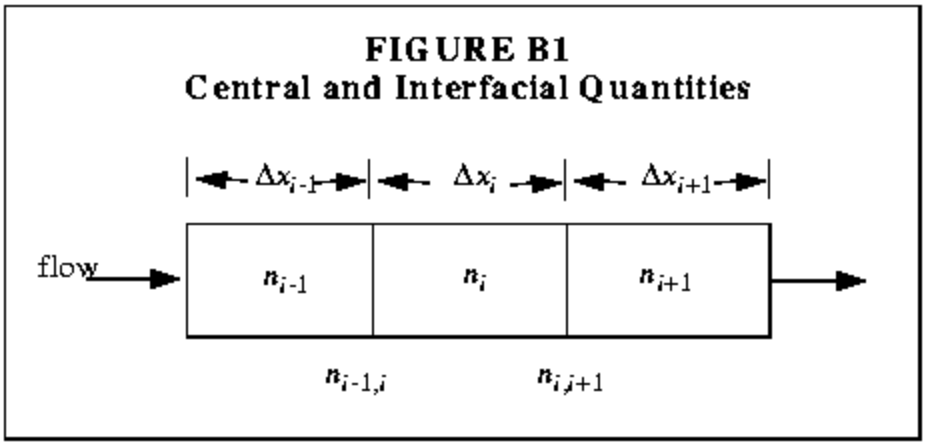
B1.0 Introduction

As discussed in Section 3.4, the spatial derivatives in the governing differential equations may be approximated using finite difference techniques. These techniques are the subject of this appendix. For a review of finite difference modeling, the reader is referred to Thomann and Mueller (1987).

We begin the discussion with a brief overview of the finite difference approximations required for the one-dimensional stream network. This overview is followed by a development of the equations for the individual segments or control-volumes that comprise the system. Two concluding sections discuss a special case of the approximations, *boundary conditions*.

B2.0 Finite Difference Equations (FDEs)

Equation (1), the mass balance equation for the stream channel, contains several spatial derivatives that may be approximated via finite differences. In general, spatial derivatives and model parameters need to be defined at both the center of the stream segment and at the segment's interfaces. These locations are defined in Figure B1, where an arbitrary quantity, *n*, is shown. In the figure, the subscript `*i*' is used to represent the quantity at the center of the segment. Quantities at the upstream interface, where segment *i-1* adjoins segment *i*, are indicated using the subscript `*i-1,i*'. In a similar fashion, the downstream interface is marked using the subscript `*i,i+1*'.



I. The Center of the Segment

The first task is to evaluate the spatial derivative, $\partial n/\partial x$, at the center of segment *i*. Using a centered difference, the spatial derivative of any variable is given by:

$$\left.\frac{\partial n}{\partial x}\right|_i = \frac{n_{i,i+1} - n_{i-1,i}}{\Delta x_i} \quad \text{(B1)}$$

Note that Equation (B1) introduces the interfacial quantities, $n_{i-1,i}$ and $n_{i,i+1}$. Since the mass balance equations were developed for the center of each segment, the interfacial quantities should be defined in terms of n_{i-1}, n_i and

n_{i+1}. To do this, we compute the interfacial quantities by interpolating linearly between the centers of the adjacent segments. This results in the following definitions:

$$n_{i-1,i} = \frac{\Delta x_i}{\Delta x_{i-1} + \Delta x_i} n_{i-1} + \frac{\Delta x_{i-1}}{\Delta x_{i-1} + \Delta x_i} n_i \quad \text{(B2)}$$

$$n_{i,i+1} = \frac{\Delta x_i}{\Delta x_{i+1} + \Delta x_i} n_{i+1} + \frac{\Delta x_{i+1}}{\Delta x_{i+1} + \Delta x_i} n_i \quad \text{(B3)}$$

II. The Segment Interfaces

We must also define the spatial derivatives at the upstream and downstream interfaces of the segment. These spatial derivative are defined as the change in *n* divided by the distance over which the change occurs. For the upstream interface, this gives:

$$\left.\frac{\partial n}{\partial x}\right|_{i-1,i} = \frac{2(n_i - n_{i-1})}{\Delta x_i + \Delta x_{i-1}} \quad \text{(B4)}$$

The same approach is used to evaluate the spatial derivative at the downstream boundary. This yields:

$$\left.\frac{\partial n}{\partial x}\right|_{i,i+1} = \frac{2(n_{i+1} - n_i)}{\Delta x_i + \Delta x_{i+1}} \quad \text{(B5)}$$

B3.0 Interior Segments - FDEs

Using the foregoing definitions, it is now possible to approximate the spatial derivatives in Equation (1). Note that the first term on the right-hand side of Equation (1) is the `*Advection*' term while the second is the `*Dispersion*' term. These two quantities are now approximated for the *interior segments* of the stream network. *Boundary segments*, i.e. the first and last segments, represent special cases that are discussed in Sections B4.0 and B5.0.

I. Advection

Using definition (B1), the advective term may be approximated as:

$$\left(-\frac{Q}{A}\frac{\partial C}{\partial x}\right)_i \cong -\frac{Q_i}{A_i}\left(\left.\frac{C_{i,i+1} - C_{i-1,i}}{\Delta x_i}\right|_i\right) \quad \text{(B6)}$$

We now replace the interfacial concentrations ($C_{i,i+1}$ and $C_{i-1,i}$) with the definitions given by Equations (B2) and (B3). This substitution results in a centered difference approximation for $\partial C/\partial x$, given equal segment lengths:

$$= -\frac{Q_i}{A_i \Delta x_i}\left(\frac{\Delta x_i}{\Delta x_{i+1} + \Delta x_i} C_{i+1} + \frac{\Delta x_{i+1}}{\Delta x_{i+1} + \Delta x_i} C_i - \frac{\Delta x_i}{\Delta x_{i-1} + \Delta x_i} C_{i-1} - \frac{\Delta x_{i-1}}{\Delta x_{i-1} + \Delta x_i} C_i\right) \quad \text{(B7)}$$

II. Dispersion

Here again we use definition (B1) to define the spatial derivative:

$$\left[\frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right)\right]_i \cong \frac{1}{A_i}\left[\frac{\left(AD\frac{\partial C}{\partial x}\right)_{i,i+1} - \left(AD\frac{\partial C}{\partial x}\right)_{i-1,i}}{\Delta x_i}\right] \quad \text{(B8)}$$

As the right side of Equation (B8) still contains spatial derivatives, we need the definitions given as Equations (B4) and (B5). Replacing the interfacial derivatives, we arrive at:

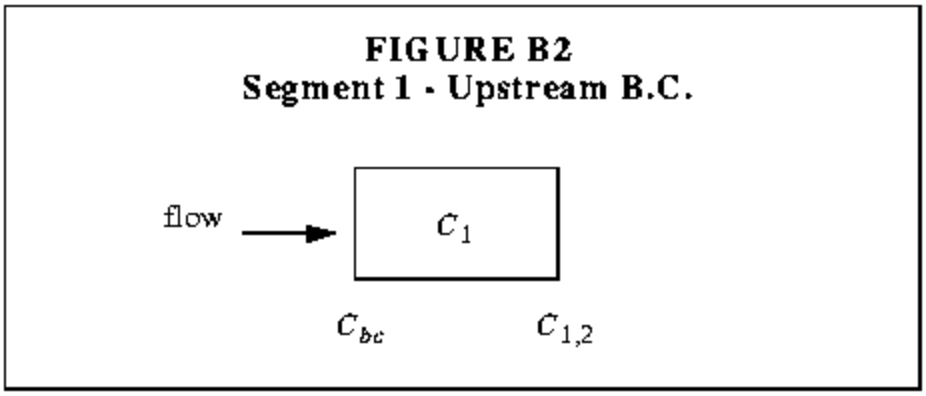
$$\cong \frac{1}{A_i \Delta x_i}\left[AD_{i,i+1}\frac{2(C_{i+1}-C_i)}{(\Delta x_i + \Delta x_{i+1})} - AD_{i-1,i}\frac{2(C_i - C_{i-1})}{(\Delta x_i + \Delta x_{i-1})}\right] \quad \text{(B9)}$$

B4.0 The First Segment - Upstream Boundary Condition

In the previous section, finite difference approximations for the spatial derivatives were developed. These approximations may be applied directly to all interior segments in the stream network. A problem arises, however, when we are dealing with the first segment, as all quantities involving the subscript *i-1* are undefined. Equations (B7) and (B9) must be modified accordingly.

I. Fixing the Upstream Boundary Condition

To effectively implement the finite difference equations for the first segment, an upstream boundary condition must be defined. This condition is shown in Figure B2, where the interfacial concentration at the upstream boundary is set to a user-supplied concentration, C_{bc}. This quantity is known as the upstream boundary condition.



II. Advection

Given the definition of C_{bc}, Equation (B6) can be modified for use in the first segment:

$$\left(-\frac{Q}{A}\frac{\partial C}{\partial x}\right)_1 = -\frac{Q_1}{A_1}\left(\frac{C_{1,2} - C_{bc}}{\Delta x_1}\right) \quad \text{(B10)}$$

As with the interior segments, replace the interfacial concentration ($C_{1,2}$) using Equation (B3):

$$= -\frac{Q_1}{A_1 \Delta x_1}\left(\frac{\Delta x_1}{\Delta x_1 + \Delta x_2}C_2 + \frac{\Delta x_2}{(\Delta x_1 + \Delta x_2)}C_1 - C_{bc}\right) \quad \text{(B11)}$$

III. Dispersion

Using the equation developed for the interior segments, the dispersive term is given by:

$$\left[\frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right)\right]_i \cong \frac{1}{A_1}\left[\frac{\left(AD\frac{\partial C}{\partial x}\right)_{1,2} - \left(AD\frac{\partial C}{\partial x}\right)_{0,1}}{\Delta x_1}\right] \quad \text{(B12)}$$

The first interfacial derivative (at 1,2) is defined as before using Equation (B5). The second term (at 0,1), requires an additional equation:

$$\left.\frac{\partial C}{\partial x}\right|_{0,1} = \frac{C_1 - C_{bc}}{\frac{1}{2}\Delta x_1} \quad \text{(B13)}$$

Note that the forward difference approximation shown above must be used in lieu of Equation (B4) to avoid introducing C_0, the concentration in the non-existent segment zero. Equations (B5) and (B13) may now be used:

$$\cong \frac{1}{A_1 \Delta x_1}\left[AD_{1,2}\frac{2(C_2 - C_1)}{\Delta x_1 + \Delta x_2} - AD_{0,1}\frac{2(C_1 - C_{bc})}{\Delta x_1}\right] \quad \text{(B14)}$$

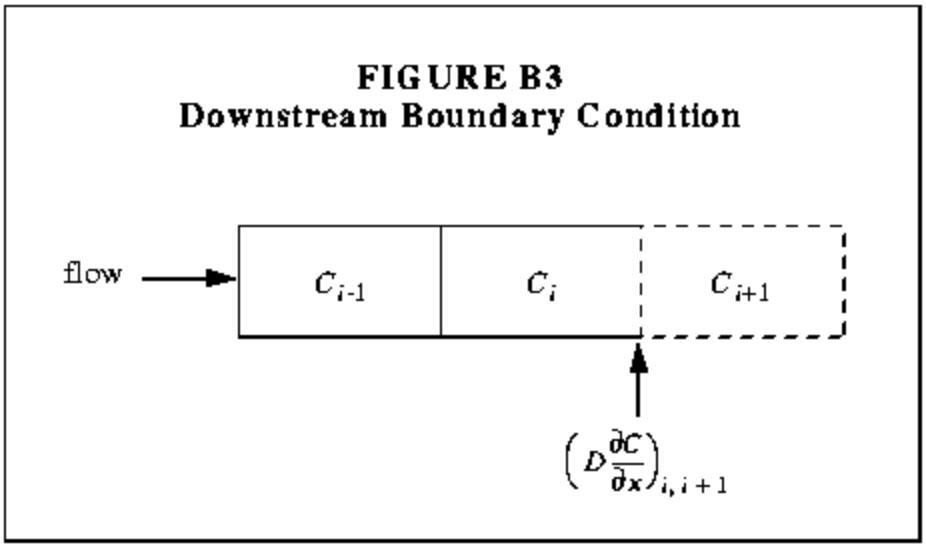
B5.0 The Last Segment - Downstream Boundary Condition

The finite difference equations developed in Section B3.0 must also be modified to accommodate the last segment in the stream network. The required modifications to Equations (B7) and (B9) are shown below.

I. Fixing the Downstream Boundary Condition

Application of Equations (B7) and (B9) to the last segment results in the introduction of an solute concentration, C_{i+1}. This concentration is undefined as segment $i+1$ does not exist. The last segment, i, and the fictitious segment, $i+1$, are shown in Figure B3.

FIGURE B3
Downstream Boundary Condition



To eliminate C_{i+1} from the finite difference equations, the dispersive flux at the interface between segments i and $i+1$ is defined as in Equation (27):

$$\left(D\frac{\partial C}{\partial x}\right)\Big|_{i,i+1} = DSBOUND \quad \text{(B15)}$$

where DSBOUND is a user-supplied value for the dispersive flux. Note that if DSBOUND is set equal to zero, Equation (B15) is a *zero gradient boundary condition* (Arden and Astill, 1970).

Using the definition of the interfacial first derivative given by Equation (B5), Equation (B15) is solved for C_{i+1}, yielding:

$$C_{i+1} = \frac{(\Delta x_i + \Delta x_{i+1})\, DSBOUND}{2D_{i,i+1}} + C_i \quad \text{(B16)}$$

Equation (B16) is now used to eliminate C_{i+1} from Equations (B7) and (B9).

II. Advection

Substituting Equation (B16) into (B7) and replacing Δx_{i+1} with Δx_i, we arrive at the finite difference approximation of the advection term:

$$\left(\frac{-Q}{A}\frac{\partial C}{\partial x}\right)_i = -\frac{Q_i}{A_i \Delta x_i}\left(\frac{\frac{1}{2}\Delta x_i DSBOUND}{D_{i,i+1}} + C_i - \frac{\Delta x_i}{(\Delta x_{i-1} + \Delta x_i)}C_{i-1} - \frac{\Delta x_{i-1}}{(\Delta x_{i-1} + \Delta x_i)}C_i\right) \quad \text{(B17)}$$

III. Dispersion

The finite difference approximation of the dispersion term is now developed by substituting Equation (B16) into (B9) and replacing Δx_{i+1} with Δx_i:

$$\left[\frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right)\right]_i = \frac{1}{A_i \Delta x_i}\left[A_{i,i+1} DSBOUND - AD_{i-1,i}\frac{2(C_i - C_{i-1})}{(\Delta x_i + \Delta x_{i-1})}\right] \quad \text{(B18)}$$

Appendix C:

Crank-Nicolson Matrix Coefficients

Numerical Solution of the Dynamic Equations

C1.0 Introduction

As discussed in Section 3.4, the Crank-Nicolson method is used to solve the dynamic solute transport equations (Equations (2) and (6)). The Crank-Nicolson solution of the main channel equation (Equation (6)) is the subject of this appendix. A similar discussion for the storage zone (Equations (2) is given in the main body of this report (Section 3.4.4).

The governing equation for the main channel is given by:

$$\frac{dC}{dt} = L[C] + \frac{q_{LIN}}{A_i}(C_L - C_i) + \alpha(C_S - C_i) + \rho\hat{\lambda}(C_{sed} - K_d C_i) - \lambda C_i \quad \text{(C1)}$$

where $L[C]$ is the finite difference approximation of the advection and dispersion terms, as given in Appendix B. In order to attain second order accuracy in both time and space, Equation (C1) is evaluated at the intermediate time level, *j+1/2*. To this end, the time derivative is approximated at the intermediate time level using a centered finite difference:

$$\left.\frac{dC}{dt}\right|_{j+\frac{1}{2}} = \frac{C_i^{j+1} - C_i^j}{\Delta t} \quad \text{(C2)}$$

where

- Δt the integration time step [T]
- *j* denotes the value of a parameter or variable at the current time
- *j+1* denotes the value of a parameter or variable at the advanced time

Furthermore, the right-hand side at *j+1/2* is simply the average of the terms evaluated at times *j* and *j+1*. Combining this average with Equation (C2), Equation (C1) becomes:

$$\frac{C_i^{j+1} - C_i^j}{\Delta t} = \frac{G[C, C_S, C_{sed}]^{j+1} + G[C, C_S, C_{sed}]^j}{2} \quad \text{(C3)}$$

where

$$G[C, C_S, C_{sed}] = L[C] + \frac{q_{LIN}}{A_i}(C_L - C_i) + \alpha(C_S - C_i) + \rho\hat{\lambda}(C_{sed} - K_d C_i) - \lambda C_i \quad \text{(C4)}$$

Because Equation (C3) is dependent on the solute concentrations in the neighboring segments at the advanced time level (C_{i-1}, C_{i+1} at time *j+1*), it is not possible to explicitly solve for C_i^{j+1}. In the sections that follow,

Equation (C3) is rearranged so that all of the known quantities (i.e. those at time *j*) appear on the right-hand side and all of the unknown quantities (i.e. those at time *j+1*) appear on the left. This rearrangement yields:

$$E_i C_{i-1}^{j+1} + F_i C_i^{j+1} + G_i C_{i+1}^{j+1} = R_i \quad \text{(C5)}$$

where *E*, *F* and *G* are matrix coefficients and *R* is a forcing function.

We now proceed to develop the matrix coefficients and the forcing function by examining each term in Equation (C3). In general, unknown quantities in each term contribute to the matrix coefficients while known quantities make up the forcing function. After going through each term, individual contributions are summed (see Section C4.0) to obtain *E*, *F*, *G* and *R*.

C2.0 Left-Hand Side Of Equation C3 - Time Derivative

As noted above, Equation (C3) must be rearranged to develop the coefficients in Equation (C5). We begin this process by considering the left side of Equation (C3), the time derivative. Leaving the unknown concentration, C_i^{j+1}, on the left side and moving the known concentration to the right side, the contributions are given by:

$$F_{dt} = \frac{1}{\Delta t} \quad \text{(C6)}$$

$$R_{dt} = \frac{1}{\Delta t} C_i^j \quad \text{(C7)}$$

C3.0 Right-Hand Side of Equation C3

The right side of Equation (C3) must also be rearranged such that the unknown quantities appear on the left side of Equation (C5) and the known quantities appear on the right. As shown in Equation (C4), $G[C, C_S, C_{sed}]$ is made up of several terms. Each term in $G[\ ]$ is examined below.

C3.1 Advection

In Equation (C4), the operator $L[C]$ represents the finite difference approximation of the spatial derivatives describing advection and dispersion. The contribution of the advection term is the subject of this section, while the dispersion term is discussed subsequently.

Using the finite difference equations for the advection term (Equations (B7), (B11), and (B17)), we develop the contributions to the matrix coefficients shown below. Note that due to the presence of boundary conditions, the contributions depend on the type of segment for which Equation (C5) is developed.

Interior Segments

Consulting Equation (B7), we see that:

$$E_{adv} = -\frac{Q_i^{j+1}}{2A_i^{j+1}(\Delta x_{i-1} + \Delta x_i)} \quad \text{(C8)}$$

$$F_{adv} = \frac{Q_i^{j+1}}{2A_i^{j+1}\Delta x_i}\left(\frac{\Delta x_{i+1}}{\Delta x_{i+1}+\Delta x_i} - \frac{\Delta x_{i-1}}{\Delta x_{i-1}+\Delta x_i}\right) \quad \text{(C9)}$$

$$G_{adv} = \frac{Q_i^{j+1}}{2A_i^{j+1}(\Delta x_{i+1}+\Delta x_i)} \quad \text{(C10)}$$

$$R_{adv} = \frac{Q_i^j}{2A_i^j}\left[\frac{1}{(\Delta x_{i-1}+\Delta x_i)}C_{i-1}^j - \frac{1}{\Delta x_i}\left(\frac{\Delta x_{i+1}}{\Delta x_{i+1}+\Delta x_i} - \frac{\Delta x_{i-1}}{\Delta x_{i-1}+\Delta x_i}\right)C_i^j - \frac{1}{(\Delta x_{i+1}+\Delta x_i)}C_{i+1}^j\right] \quad \text{(C11)}$$

First Segment

Using Equation (B11):

$$F_{adv} = \frac{Q_1^{j+1}\Delta x_2}{2A_1^{j+1}\Delta x_1(\Delta x_1+\Delta x_2)} \quad \text{(C12)}$$

$$G_{adv} = \frac{Q_1^{j+1}}{2A_1^{j+1}(\Delta x_1+\Delta x_2)} \quad \text{(C13)}$$

$$R_{adv} = -\frac{Q_1^j\Delta x_2}{2A_1^j\Delta x_1(\Delta x_1+\Delta x_2)}C_1^j - \frac{Q_1^j}{2A_1^j(\Delta x_1+\Delta x_2)}C_2^j + \frac{1}{2}\left(\frac{Q_1^j}{A_1^j\Delta x_1}C_{bc}^j + \frac{Q_1^{j+1}}{A_1^{j+1}\Delta x_1}C_{bc}^{j+1}\right) \quad \text{(C14)}$$

Last Segment

Evaluating Equation (B17) at *j+1/2* yields:

$$E_{adv} = -\frac{Q_i^{j+1}}{2A_i^{j+1}(\Delta x_{i-1}+\Delta x_i)} \quad \text{(C15)}$$

$$F_{adv} = \frac{Q_i^{j+1}}{2A_i^{j+1}\Delta x_i}\left(1 - \frac{\Delta x_{i-1}}{\Delta x_{i-1}+\Delta x_i}\right) \quad \text{(C16)}$$

$$R_{adv} = \frac{Q_i^j}{2A_i^j(\Delta x_{i-1}+\Delta x_i)}C_{i-1}^j - \frac{Q_i^j}{2A_i^j\Delta x_i}\left(1 - \frac{\Delta x_{i-1}}{\Delta x_{i-1}+\Delta x_i}\right)C_i^j - \frac{DSBOUND}{4D_{i,i+1}}\left(\frac{Q_i^j}{A_i^j} + \frac{Q_i^{j+1}}{A_i^{j+1}}\right) \quad \text{(C17)}$$

C3.2 Dispersion

As with the advection term, the finite difference equations given in Appendix B (Equations (B9), (B14) and (B18)) are used to determine the contributions to the matrix coefficients.

Interior Segments

Equation (B9), evaluated at *j+1/2*, yields:

$$E_{disp} = -\frac{(AD)_{i-1,i}^{j+1}}{A_i^{j+1}\Delta x_i(\Delta x_i+\Delta x_{i-1})} \quad \text{(C18)}$$

$$F_{disp} = \frac{1}{A_i^{j+1}\Delta x_i}\left(\frac{(AD)_{i,i+1}^{j+1}}{\Delta x_i + \Delta x_{i+1}} + \frac{(AD)_{i-1,i}^{j+1}}{\Delta x_i + \Delta x_{i-1}}\right) \quad \text{(C19)}$$

$$G_{disp} = -\frac{(AD)_{i,i+1}^{j+1}}{A_i^{j+1}\Delta x_i(\Delta x_i + \Delta x_{i+1})} \quad \text{(C20)}$$

$$R_{disp} = \frac{1}{A_i^j \Delta x_i}\left[\frac{(AD)_{i-1,i}^{j}}{(\Delta x_i + \Delta x_{i-1})}C_{i-1}^j - \left(\frac{(AD)_{i,i+1}^{j}}{\Delta x_i + \Delta x_{i+1}} + \frac{(AD)_{i-1,i}^{j}}{\Delta x_i + \Delta x_{i-1}}\right)C_i^j + \frac{(AD)_{i,i+1}^{j}}{(\Delta x_i + \Delta x_{i+1})}C_{i+1}^j\right] \quad \text{(C21)}$$

First Segment

Using the finite difference equation for the first segment (Equation (B14)):

$$F_{disp} = \frac{1}{A_1^{j+1}\Delta x_1}\left(\frac{(AD)_{1,2}^{j+1}}{\Delta x_1 + \Delta x_2} + \frac{(AD)_{0,1}^{j+1}}{\Delta x_1}\right) \quad \text{(C22)}$$

$$G_{disp} = -\frac{(AD)_{1,2}^{j+1}}{A_1^{j+1}\Delta x_1(\Delta x_1 + \Delta x_2)} \quad \text{(C23)}$$

$$\begin{aligned} R_{disp} = &-\frac{1}{A_1^j \Delta x_1}\left(\frac{(AD)_{1,2}^{j}}{\Delta x_1 + \Delta x_2} + \frac{(AD)_{0,1}^{j}}{\Delta x_1}\right)C_1^j \\ &+ \frac{(AD)_{1,2}^{j}}{A_1^j \Delta x_1(\Delta x_1 + \Delta x_2)}C_2^j + \frac{1}{\Delta x_1^2}\left(\frac{(AD)_{0,1}^{j}}{A_1^j}C_{bc}^j + \frac{(AD)_{0,1}^{j+1}}{A_1^{j+1}}C_{bc}^{j+1}\right) \end{aligned} \quad \text{(C24)}$$

Last Segment

Finally, Equation (B18) at *j+1/2* yields:

$$E_{disp} = -\frac{(AD)_{i-1,i}^{j+1}}{A_i^{j+1}\Delta x_i(\Delta x_i + \Delta x_{i-1})} \quad \text{(C25)}$$

$$F_{disp} = \frac{(AD)_{i-1,i}^{j+1}}{A_i^{j+1}\Delta x_i(\Delta x_i + \Delta x_{i-1})} \quad \text{(C26)}$$

$$R_{disp} = \frac{(AD)_{i-1,i}^{j}}{A_i^j \Delta x_i(\Delta x_i + \Delta x_{i-1})}C_{i-1}^j - \frac{(AD)_{i-1,i}^{j}}{A_i^j \Delta x_i(\Delta x_i + \Delta x_{i-1})}C_i^j + \frac{DSBOUND}{2\Delta x_i}\left(\frac{A_{i,i+1}^j}{A_i^j} + \frac{A_{i,i+1}^{j+1}}{A_i^{j+1}}\right) \quad \text{(C27)}$$

C3.3 Lateral Inflow

The third term in *G*[ ], lateral inflow, is also evaluated at *j+1/2*:

$$\left(\frac{q_{LIN}}{A}(C_L - C)\right)^{j+\frac{1}{2}} = \frac{\left(\frac{q_{LIN}}{A}(C_L - C_i)\right)^j + \left(\frac{q_{LIN}}{A}(C_L - C_i)\right)^{j+1}}{2} \quad \text{(C28)}$$

The contributions are thus:

$$F_{lat} = \frac{q_{LIN}^{j+1}}{2A_i^{j+1}} \quad \text{(C29)}$$

$$R_{lat} = \frac{1}{2}\left(\frac{q_{LIN}^{j}}{A_i^{j}} C_L^{j} + \frac{q_{LIN}^{j+1}}{A_i^{j+1}} C_L^{j+1} \right) - \frac{q_{LIN}^{j}}{2A_i^{j}} C_i^{j} \quad \text{(C30)}$$

C3.4 Transient Storage

The transient storage term is now evaluated at *j+1/2*:

$$\alpha (C_S - C_i)^{j+\frac{1}{2}} = \frac{\alpha (C_S - C_i)^{j} + \alpha (C_S - C_i)^{j+1}}{2} = \frac{\alpha}{2}\left(C_S^{j} - C_i^{j} + C_S^{j+1} - C_i^{j+1} \right) \quad \text{(C31)}$$

To decouple the governing transport equations (see Section 3.4.5), Equation (19) is used to eliminate C_S^{j+1}. Equation (C31) thus becomes:

$$= \frac{\alpha}{2}\left\{ C_S^{j} - C_i^{j} - C_i^{j+1} + \frac{1}{2 + \gamma^{j+1} + \Delta t\hat{\lambda}_S + \Delta t\lambda_S}\left[\left(2 - \gamma^{j} - \Delta t\hat{\lambda}_S - \Delta t\lambda_S \right) C_S^{j} + \gamma^{j} C_i^{j} + \gamma^{j+1} C_i^{j+1} + 2\Delta t\hat{\lambda}_S \hat{C}_S \right] \right\} \quad \text{(C32)}$$

The contributions to the matrix coefficients are given by:

$$F_{stor} = \frac{\alpha}{2}\left(1 - \frac{\gamma^{j+1}}{2 + \gamma^{j+1} + \Delta t\hat{\lambda}_S + \Delta t\lambda_S} \right) \quad \text{(C33)}$$

$$R_{stor} = \frac{\alpha}{2}\left[\left(1 + \frac{2 - \gamma^{j} - \Delta t\hat{\lambda}_S - \Delta t\lambda_S}{2 + \gamma^{j+1} + \Delta t\hat{\lambda}_S + \Delta t\lambda_S} \right) C_S^{j} + \left(\frac{\gamma^{j}}{2 + \gamma^{j+1} + \Delta t\hat{\lambda}_S + \Delta t\lambda_S} - 1 \right) C_i^{j} \right] + \frac{\alpha \Delta t\hat{\lambda}_S \hat{C}_S}{2 + \gamma^{j+1} + \Delta t\hat{\lambda}_S + \Delta t\lambda_S} \quad \text{(C34)}$$

C3.5 Sorption

The sorption term is evaluated at *j+1/2*:

$$\rho\hat{\lambda} (C_{sed} - K_d C_i)^{j+\frac{1}{2}} = \frac{\rho\hat{\lambda}}{2}\left(C_{sed}^{j} - K_d C_i^{j} + C_{sed}^{j+1} - K_d C_i^{j+1} \right) \quad \text{(C35)}$$

To decouple the governing transport equations (see Section 2.4.5), the Crank-Nicolson solution of the streambed sediment equation is used to eliminate C_{sed}^{j+1}. Equation (C35) thus becomes:

$$= \frac{\rho\hat{\lambda}}{2}\left(C_{sed}^{j} - K_d C_i^{j} - K_d C_i^{j+1} + \frac{2 - \Delta t\hat{\lambda}}{2 + \Delta t\hat{\lambda}} C_{sed}^{j} + \frac{\Delta t\hat{\lambda} K_d}{2 + \Delta t\hat{\lambda}} C_i^{j} + \frac{\Delta t\hat{\lambda} K_d}{2 + \Delta t\hat{\lambda}} C_i^{j+1} \right) \quad \text{(C36)}$$

The contributions to the matrix coefficients are given by:

$$F_{sorp} = \frac{\rho\hat{\lambda}}{2}\left(K_d - \frac{\Delta t\hat{\lambda}K_d}{2+\Delta t\hat{\lambda}}\right) \quad \text{(C37)}$$

$$R_{sorp} = -F_{sorp}C_i^j + \frac{\rho\hat{\lambda}}{2}\left(1 + \frac{2-\Delta t\hat{\lambda}}{2+\Delta t\hat{\lambda}}\right)C_{sed}^j \quad \text{(C38)}$$

C3.6 First-Order Decay

The final term in $G[\ ]$ represents first-order decay or production. This term is evaluated at $j+1/2$:

$$(-\lambda C)^{j+\frac{1}{2}} = \frac{-(\lambda C)^j - (\lambda C)^{j+1}}{2} = -\frac{\lambda}{2}\left(C_i^j + C_i^{j+1}\right) \quad \text{(C39)}$$

The contributions are given by:

$$F_{decay} = \frac{\lambda}{2} \quad \text{(C40)}$$

$$R_{decay} = -F_{decay}C_i^j \quad \text{(C41)}$$

C4.0 Matrix Coefficients

The matrix coefficients (E, F, G) and the forcing function (R) from Equation (C5) may now be developed by summing the contributions given in Sections C2.0 and C3.0, and multiplying by Δt:

$$E_i = \Delta t\,(E_{adv} + E_{disp}) \quad \text{(C42)}$$

$$F_i = \Delta t\,(F_{dt} + F_{adv} + F_{disp} + F_{lat} + F_{stor} + F_{sorp} + F_{decay}) \quad \text{(C43)}$$

$$G_i = \Delta t\,(G_{adv} + G_{disp}) \quad \text{(C44)}$$

$$R_i = \Delta t\,(R_{dt} + R_{adv} + R_{disp} + R_{lat} + R_{stor} + R_{sorp} + R_{decay}) \quad \text{(C45)}$$

Appendix D:

Steady State Matrix Coefficients

The purpose of this appendix is to develop the matrix coefficients (E, F, G) and the forcing function (R) from Equation (26). In the text that follows, the contributions to E, F, G and R are given on a term-by-term basis. The final values are determined by summing the individual contributions.

D1.0 Advection

Equations (B7), (B11), and (B17) are used to approximate the spatial derivative describing advective transport. These equations contribute to the matrix coefficients as detailed below. Due to boundary conditions, the contributions depend on the type of segment for which Equation (26) is developed.

Interior Segments

$$E_{adv} = -\frac{Q_i}{A_i}\left(\frac{1}{\Delta x_{i-1} + \Delta x_i}\right) \quad \text{(D1)}$$

$$F_{adv} = \frac{Q_i}{A_i \Delta x_i}\left(\frac{\Delta x_{i+1}}{\Delta x_i + \Delta x_{i+1}} - \frac{\Delta x_{i-1}}{\Delta x_i + \Delta x_{i-1}}\right) \quad \text{(D2)}$$

$$G_{adv} = \frac{Q_i}{A_i}\left(\frac{1}{\Delta x_i + \Delta x_{i+1}}\right) \quad \text{(D3)}$$

First Segment

$$F_{adv} = \frac{Q_1}{A_1 \Delta x_1}\left(\frac{\Delta x_2}{\Delta x_1 + \Delta x_2}\right) \quad \text{(D4)}$$

$$G_{adv} = \frac{Q_1}{A_1}\left(\frac{1}{\Delta x_1 + \Delta x_2}\right) \quad \text{(D5)}$$

$$R_{adv} = \frac{Q_1}{A_1}\frac{C_{bc}}{\Delta x_1} \quad \text{(D6)}$$

Last Segment

$$E_{adv} = -\frac{Q_i}{A_i}\left(\frac{1}{\Delta x_i + \Delta x_{i-1}}\right) \quad \text{(D7)}$$

$$F_{adv} = \frac{Q_i}{A_i \Delta x_i}\left(1 - \frac{\Delta x_{i-1}}{\Delta x_i + \Delta x_{i-1}}\right) \quad \text{(D8)}$$

$$R_{adv} = -\frac{Q_i}{A_i}\frac{DSBOUND}{2D_{i,i+1}} \quad \text{(D9)}$$

D2.0 Dispersion

As with the advection term, the finite difference equations given in Appendix B (Equations (B9), (B14) and (B18)) are used to determine the contributions to the matrix coefficients.

Interior Segments

$$E_{disp} = -\frac{2(AD)_{i-1,i}}{A_i\Delta x_i(\Delta x_i + \Delta x_{i-1})} \quad \text{(D10)}$$

$$F_{disp} = \frac{2}{A_i\Delta x_i}\left[\frac{AD_{i,i+1}}{\Delta x_i + \Delta x_{i+1}} + \frac{AD_{i-1,i}}{\Delta x_i + \Delta x_{i-1}}\right] \quad \text{(D11)}$$

$$G_{disp} = -\frac{2AD_{i,i+1}}{A_i\Delta x_i(\Delta x_i + \Delta x_{i+1})} \quad \text{(D12)}$$

First Segment

$$F_{disp} = \frac{2}{A_1\Delta x_1}\left(\frac{AD_{1,2}}{\Delta x_1 + \Delta x_2} + \frac{AD_{0,1}}{\Delta x_1}\right) \quad \text{(D13)}$$

$$G_{disp} = -\frac{2AD_{1,2}}{A_1\Delta x_1(\Delta x_1 + \Delta x_2)} \quad \text{(D14)}$$

$$R_{disp} = \frac{1}{A_1}\frac{2AD_{0,1}C_{bc}}{\Delta x_1^2} \quad \text{(D15)}$$

Last Segment

$$E_{disp} = -\frac{2AD_{i-1,i}}{A_i\Delta x_i(\Delta x_i + \Delta x_{i-1})} \quad \text{(D16)}$$

$$F_{disp} = \frac{2AD_{i-1,i}}{A_i\Delta x_i(\Delta x_i + \Delta x_{i-1})} \quad \text{(D17)}$$

$$R_{disp} = \frac{A_{i,i+1}DSBOUND}{A_i\Delta x_i} \quad \text{(D18)}$$

D3.0 Lateral Inflow

$$F_{lat} = \frac{q_{LIN}}{A_i} \quad \text{(D19)}$$

$$R_{lat} = \frac{q_{LIN}}{A_i} C_L \quad \text{(D20)}$$

D4.0 Transient Storage

$$F_{stor} = \alpha A_S \left(\frac{\lambda_S + \hat{\lambda}_S}{\alpha A + A_S \lambda_S + A_S \hat{\lambda}_S} \right) \quad \text{(D21)}$$

$$R_{stor} = \frac{\alpha A_S \hat{\lambda}_S \hat{C}_S}{\alpha A + A_S \lambda_S + A_S \hat{\lambda}_S} \quad \text{(D22)}$$

D5.0 Decay

$$F_{decay} = \lambda \quad \text{(D23)}$$

D6.0 Matrix Coefficients

The matrix coefficients (E, F, G) and the forcing function (R) from Equation (26) may now be developed by summing the contributions given in Sections D1.0 through D5.0;

$$E_i = E_{adv} + E_{disp} \quad \text{(D24)}$$

$$F_i = F_{adv} + F_{disp} + F_{lat} + F_{stor} + F_{decay} \quad \text{(D25)}$$

$$G_i = G_{adv} + G_{disp} \quad \text{(D26)}$$

$$R_i = R_{adv} + R_{disp} + R_{lat} \quad \text{(D27)}$$
